

38. (b) After first bounce, height of ball

$$= \left(\frac{2}{3}\right)^1 \times 36$$

and after third bounce, height of ball

$$= \left(\frac{2}{3}\right)^3 \times 36$$

$$= \frac{8}{27} \times 36 = \frac{8 \times 4}{3} = \frac{32}{3} \text{ m} = 10\frac{2}{3} \text{ m}$$

Hence, the required height at third bounce is $10\frac{2}{3}$ m.

39. (c) Product of four consecutive numbers plus one is always a square.

Illustration 1 Let four consecutive numbers be 3, 4, 5 and 6.

$$\therefore (3 \times 4 \times 5 \times 6) + 1 = 361 = (19)^2$$

Illustration 2 Let four consecutive numbers be 9, 10, 11 and 12.

$$\therefore (9 \times 10 \times 11 \times 12) + 1 = 11881 = (109)^2$$

40. (b) The difference of two consecutive cubes is never divisible by 2.

Illustration 1 Let the two consecutive numbers be 4 and 5.

$$\therefore (5)^3 - (4)^3 = 125 - 64 = 61$$

Illustration 2 Let the two consecutive numbers be 9 and 10.

$$\therefore (10)^3 - (9)^3 = 1000 - 729 = 271$$

41. (b) $\sqrt{\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]}$

$$= \sqrt{\frac{(0.75)^3}{0.25} + [1.75 + (0.75)^2]}$$

$$= \sqrt{3 \times (0.75)^2 + 1.75 + (0.75)^2}$$

$$= \sqrt{4 \times (0.75)^2 + 1.75} = \sqrt{4} = 2$$

Hence, the required square root is 2.

42. (d) $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$

$$= \frac{5 + \sqrt{10}}{5\sqrt{5} - 4\sqrt{5} - 4\sqrt{2} + 5\sqrt{2}}$$

$$= \frac{(5 + \sqrt{10}) \times (\sqrt{5} - \sqrt{2})}{(\sqrt{5} + \sqrt{2}) \times (\sqrt{5} - \sqrt{2})}$$

$$= \frac{5\sqrt{5} - 5\sqrt{2} + \sqrt{50} - \sqrt{20}}{\sqrt{5^2 - 2^2}} \quad [\text{by rationalisation}]$$

$$[\because (a-b)(a+b) = a^2 - b^2]$$

$$= \frac{5\sqrt{5} - 5\sqrt{2} + 5\sqrt{2} - 2\sqrt{5}}{5 - 2} = \frac{3\sqrt{5}}{3} = \sqrt{5}$$

43. (a) We have, $x = \frac{91}{216}$

$$\Rightarrow 1 - x = 1 - \frac{91}{216} = \frac{125}{216} = \left(\frac{5}{6}\right)^3$$

$$\Rightarrow (1-x)^{\frac{1}{3}} = \frac{5}{6} \Rightarrow \frac{1}{(1-x)^{\frac{1}{3}}} = \frac{6}{5}$$

$$\Rightarrow 3 - \frac{1}{(1-x)^{\frac{1}{3}}} = 3 - \frac{6}{5} = \frac{9}{5}$$

44. (b) We have, $\sqrt{2}, 6^{1/4}, 4^{1/3}$

The LCM of 2, 4, 3 are 12.

$$(2^6)^{\frac{1}{12}}; (6^3)^{\frac{1}{12}}; (4^4)^{\frac{1}{12}}$$

$$(64)^{\frac{1}{12}}; (216)^{\frac{1}{12}}; (256)^{\frac{1}{12}}$$

$$\sqrt{2} < 6^{\frac{1}{4}} < 4^{\frac{1}{3}} \text{ or } \sqrt{2} > \sqrt[4]{6} > \sqrt[3]{4}$$

45. (b) We have, $x = \sqrt{3} + \sqrt{2}$

$$\therefore \frac{1}{x} = \frac{1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}}$$

$$= \frac{\sqrt{3} - \sqrt{2}}{3 - 2} = \sqrt{3} - \sqrt{2}$$

$$\therefore x + \frac{1}{x} = 2\sqrt{3}$$

On cubing both sides, we get

$$x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) = 24\sqrt{3}$$

$$\therefore x^3 + \frac{1}{x^3} + x + \frac{1}{x} = 24\sqrt{3} - 2\left(x + \frac{1}{x}\right)$$

$$= 24\sqrt{3} - 2 \times 2\sqrt{3}$$

$$= 24\sqrt{3} - 4\sqrt{3} = 20\sqrt{3}$$

46. (a) Given, $x = 2^{1/3} + 2^{-1/3}$

$$\therefore 2x^3 - 6x - 5 = 2(2^{1/3} + 2^{-1/3})^3 - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 2[2 + 2^{-1} + 3(2^{1/3} + 2^{-1/3})] - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 4 + 2 \times \frac{1}{2} + 6(2^{1/3} + 2^{-1/3}) - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 4 + 1 - 5 = 0$$

47. (b) We have, $4^x 2^y = 128$

$$\Rightarrow 2^{2x} 2^y = (2)^7$$

$$\Rightarrow 2^{2x+y} = (2)^7$$

$$\Rightarrow 2x + y = 7$$

$$\Rightarrow y = 7 - 2x \quad \dots(i)$$

$$\text{Now, } 3^{3x} \cdot 3^{2y} = 9^{xy}$$

$$\Rightarrow 3^{3x+2y} = 3^{2xy}$$

$$\Rightarrow 3x + 2y = 2xy \quad \dots(ii)$$

$$\Rightarrow 3x + 2(7 - 2x) = 2x(7 - 2x)$$

[from Eq. (i)]

$$\Rightarrow 3x + 14 - 4x = 14x - 4x^2$$

$$\Rightarrow 4x^2 - 15x + 14 = 0$$

$$\Rightarrow 4x^2 - 8x - 7x + 14 = 0$$

$$\Rightarrow 4x(x - 2) - 7(x - 2) = 0$$

$$\Rightarrow (4x - 7)(x - 2) = 0$$

$$\Rightarrow x = \frac{7}{4} \text{ or } x = 2$$

$$\therefore y = \frac{7}{2} \text{ or } y = 3$$

$$\therefore x + y = \frac{21}{4} \text{ or } x + y = 5$$

48. (a) Refer to question 32.

06

TIME AND DISTANCE

Regularly (4-7) questions have been asked from this chapter. Generally the type of questions vary from simple to quite complex. But if you know the basics, then this section becomes easy to score area.



Distance

The length of the path travelled by any object or a person between two places is known as distance. The unit of distance is m or km.

Time

The duration in hours, minutes or seconds spent to cover a certain distance is called time.

Speed

The distance travelled by any object or a person per unit time is known as speed of that object or a person. The unit of speed is m/s or km/h.

So,
$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

EXAMPLE 1. There are 20 poles with a constant distance between each pole. A car takes 24 s to reach the 12th pole. How much time will it take to reach the last pole?

- a. 25.25 s b. 17.45 s c. 35.75 s d. 41.45 s

Sol. d. Suppose distance between each pole is 1m.

∴ Total distance = 19 m

It takes 24 s to cover 11 m.

∴ To cover 19 m, it will take $\frac{24}{11} \times 19 = 41.45$.

Conversion of Unit

- (i) To convert speed of an object from km/h to m/s, multiply it by $\frac{5}{18}$.

e.g. $36 \text{ km/h} = 36 \times \frac{5}{18} \text{ m/s} = 10 \text{ m/s}$

- (ii) To convert speed of an object from m/s to km/h, multiply it by $\frac{18}{5}$.

e.g. $5 \text{ m/s} = 5 \times \frac{18}{5} \text{ km/h} = 18 \text{ km/h}$

Average Speed

The average speed of an object is defined as total distance travelled divided by total time taken.

i.e. Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

EXAMPLE 2. A man completes 30 km of a journey at 6 km/h and the remaining 40 km of the journey in 5 h. Then, his average speed for the whole journey is

- a. 5 km/h b. 7 km/h
c. 7.5 km/h d. None of these

Sol. b. Average speed = $\frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{70}{10} = 7 \text{ km/h}$

Relative Speed

The speed of an object with respect to other is called relative speed. Suppose two bodies are moving with speeds of $x \text{ km/h}$ and $y \text{ km/h}$ respectively, then their relative speed will be

- (i) $(x + y) \text{ km/h}$, if both the bodies are moving in opposite directions.
(ii) $(x - y) \text{ km/h}$, if both the bodies are moving in same direction.

EXAMPLE 3. Two persons 27 km apart setting out at the same time are together in 9 h, if they walk in the same direction but in 3 h, if they walk in opposite directions. Then, their rates of walking (speeds) are

- a. 2 km/h and 4 km/h b. 3 km/h and 5 km/h
c. 4 km/h and 8 km/h d. None of these

Sol. d. Let the first person be walking faster with speed $x \text{ km/h}$ and second walking with speed $y \text{ km/h}$.

Case I Both walking in same directions.

$\therefore$ Distance travelled by first person in 9 h = $9x \text{ km}$
and distance travelled by second person in 9 h = $9y \text{ km}$
As both are 27 km apart

$\therefore 9x - 9y = 27 \Rightarrow x - y = 3 \quad \dots(i)$

Case II Both walking in opposite directions.

$\therefore$ Distance travelled by first person in 3 h = $3x$
and distance travelled by second person in 3 h = $3y$
So, by condition, $3x + 3y = 27 \Rightarrow x + y = 9 \quad \dots(ii)$

On adding Eqs. (i) and (ii), we get

$\Rightarrow 2x = 12 \Rightarrow x = 6 \text{ km/h}$

Put the value of x in Eq. (ii), we get

$6 + y = 9 \Rightarrow y = 3 \text{ km/h}$

So, their speeds are 6 km/h and 3 km/h.

Important Rules and Formulae

Rule 1 If A travels with speed $x \text{ km/h}$ for $t_1 \text{ h}$ and with speed $y \text{ km/h}$ for $t_2 \text{ h}$, then average speed = $\frac{xt_1 + yt_2}{t_1 + t_2}$

EXAMPLE 4. A man walks at the rate of 5 km/h for 6 h and at 4 km/h for 12 h. Find out the average speed (in km/h) of the man.

- a. $2\frac{1}{3}$ b. $5\frac{1}{3}$ c. $7\frac{1}{3}$ d. None of these

Sol. d. Here, $x = 5 \text{ km/h}$, $y = 4 \text{ km/h}$, $t_1 = 6 \text{ h}$ and $t_2 = 12 \text{ h}$

$\therefore$ Average speed = $\frac{xt_1 + yt_2}{t_1 + t_2} = \frac{5 \times 6 + 4 \times 12}{6 + 12} = \frac{30 + 48}{18}$
 $= \frac{78}{18} = 4\frac{1}{3} \text{ km/h}$

Rule 2 If a man/vehicle covers two equal distances with the speed of $x \text{ km/h}$ and $y \text{ km/h}$ respectively, then the average speed of the man/vehicle for complete journey will be $\frac{2xy}{x + y}$.

EXAMPLE 5. A man covers half of his journey at 6 km/h and the remaining half at 3 km/h. Then, his average speed is

- a. 1 km/h b. 2 km/h c. 3 km/h d. 4 km/h

Sol. d. Here, $x = 6 \text{ km/h}$ and $y = 3 \text{ km/h}$

$\therefore$ Average speed = $\frac{2xy}{x + y} = \frac{2 \times 6 \times 3}{6 + 3} = \frac{36}{9} = 4 \text{ km/h}$

Rule 3 If a man changes his speed to $\left(\frac{x}{y}\right)$ of his usual

speed and gets late by $t \text{ min}$, then the usual time taken by him = $\frac{t \times x}{(y - x)}$.

Note If a person changes his speed to $\left(\frac{x}{y}\right)$ of his usual and reaches early by $t \text{ min}$, then usual time taken by him = $\frac{t \times x}{(x - y)}$.

EXAMPLE 6. If a man travels with a speed of $\frac{2}{5}$ times of his original speed and he reached his office 15 min late to fixed time, then the time taken with his original speed, is

- a. 10 min b. 15 min c. 20 min d. 25 min

Sol. a. Here, $x = 2$, $y = 5$ and $t = 15 \text{ min}$

$\therefore$ Required time = $\frac{x \times t}{y - x} = \frac{2 \times 15}{5 - 2}$
 $= \frac{2 \times 15}{3} = 2 \times 5 = 10 \text{ min}$

Rule 4 When a man travels from A to B with a speed x km/h and reaches t_1 h later after the fixed time and when he travels with a speed y km/h from A to B , he reaches his destination t_2 h before the fixed time, then distance between A and $B = \frac{xy(t_1 + t_2)}{y - x}$ km.

EXAMPLE 7. Walking at 3 km/h, Rajeev reaches his school 5 min late, if he walks at 4 km/h, he will be 5 min early. The distance of Rajeev's school from his house is

- a. $1\frac{1}{2}$ km b. 2 km c. $2\frac{1}{2}$ km d. 5 km

Sol. b. Here, $x = 3$ km/h, $y = 4$ km/h, $t_1 = \frac{5}{60}$ h and $t_2 = \frac{5}{60}$ h

$$\therefore \text{Required distance} = \frac{xy(t_1 + t_2)}{y - x} = \frac{3 \times 4 \left(\frac{5}{60} + \frac{5}{60} \right)}{4 - 3} = 3 \times 4 \times \frac{10}{60} = 2 \text{ km}$$

Rule 5 A person travels from A to B with a speed x km/h and reaches t_1 h late. Later he increases his speed by y km/h to cover the same distance and he still gets late by t_2 h, then the distance between A and $B = (t_1 - t_2)(x + y) \frac{x}{y}$.

EXAMPLE 8. A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increased his speed by 5 km/h but still he is late by 10 min. What is the distance of the school from his home?

- a. $\frac{100}{3}$ km b. $\frac{50}{3}$ km c. $\frac{100}{7}$ km d. $\frac{2}{5}$ km

Sol. a. Given, $x = 20$ km/h, $y = 5$ km/h, $t_1 = 30 \text{ min} = \frac{1}{2}$ h

$$\text{and } t_2 = 10 \text{ min} = \frac{1}{6} \text{ h}$$

$$\therefore \text{Required distance} = (t_1 - t_2)(x + y) \frac{x}{y} = \left(\frac{1}{2} - \frac{1}{6} \right) (20 + 5) \times \frac{20}{5} = \frac{100}{3} \text{ km}$$

Problem Based on Trains

Problems based on trains are same as the problems related to speed, Time and Distance. The only difference is that the length of the moving object (train) is taken into consideration in these types of problems.

Rule 6 (i) Time taken by a train x m long in passing a single post or pole or standing man = Time taken by the train to cover x m.

(ii) Time taken by a train x m long in passing an object of length y m = Time taken by the train to cover $(x + y)$ m.

EXAMPLE 9. A 100 m long train is moving at a speed of 60 km/h. In what time will it cross a signal pole?

- a. 6 s b. 12 s c. 15 s d. None of these

Sol. a. Here, speed of train = 60 km/h = $60 \times \frac{5}{18} = \frac{50}{3}$ m/s

and distance covered by train in passing a pole = Length of the train

$\therefore$ Length of train = 100 m

$$\therefore \text{Time taken to pass the pole} = \frac{\text{Distance travelled by train}}{\text{Speed of train}} = 6 \text{ s}$$

Rule 7 Time taken by a train of length l_1 metre moving at a speed of s_1 m/s to cross another train of length l_2 moving at a speed of s_2 m/s in same/opposite direction is $\frac{l_1 + l_2}{s_1 \pm s_2}$

Use '+' for opposite direction.

Use '-' for same direction.

EXAMPLE 10. Two trains of lengths 110 m and 130 m travel on parallel track. If they move in the same direction, the first one which is faster takes one minute to pass the other one completely. If they move in opposite directions then they pass each other in 3s, then the speed of the trains is

- a. 41 m/s and 39 m/s b. 32 m/s and 43 m/s
c. 42 m/s and 38 m/s d. None of these

Sol. c. Let v_1 be the velocity of faster train and v_2 be the velocity of slower train.

Case I If they move in the same direction, then

$$\frac{110 + 130}{v_1 - v_2} = 60 \text{ s} \Rightarrow v_1 - v_2 = 4 \quad \dots(i)$$

Case II If they move in opposite directions,

$$\frac{110 + 130}{v_1 + v_2} = 3 \Rightarrow v_1 + v_2 = 80 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get $2v_1 = 84 \Rightarrow v_1 = 42$ m/s

Now, putting the value of v_1 in Eq. (ii), we get

$$42 + v_2 = 80, v_2 = 38 \text{ m/s} = 3 \times 4 \times \frac{10}{60} = 2 \text{ km}$$

Rule 8 If two trains start at the same time from points A and B towards each other and after crossing each other, they take t_1 and t_2 time in reaching points B and A respectively, then ratio of their speed = $\sqrt{\frac{t_2}{t_1}}$

or
$$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

EXAMPLE 11. Two trains, one from Howrah to Patna and the other from Patna to Howrah, start simultaneously. After they meet, the trains reach their destinations after 9 h and 16 h, respectively. What is the ratio of their speeds?

- a. 4:3 b. 3:1
c. 4:5 d. 3:2

Sol. a. Let us name the trains as A and B . Then, A 's speed : B 's speed = $\sqrt{t_2} : \sqrt{t_1} = \sqrt{16} : \sqrt{9} = 4 : 3$.

Problems Based on Boat and Stream

Boats and streams is an application of speed, time and distance. Some of the important terms are explained below

- **Still water** If the speed of water is zero, then water is considered to be still water.
- **Stream water** If the water of a river is moving at a certain speed, then it is called stream water.
- **Downstream motion** In water, the direction along the stream is called downstream.
- **Upstream motion** In water, the direction against the stream is called upstream.

Some rules and problems related to boat and stream are given below

Rule 9 If the speed of a boat in still water be x km/h and speed of stream be y km/h, then

- (i) Speed of boat downstream = $(x + y)$ km/h
(ii) Speed of boat upstream = $(x - y)$ km/h

EXAMPLE 12. A sailor goes 8 km downstream in 40 min and returns back in 1 h. Then, the speed of the sailor in still water and the speed of the current is

- a. 5 km/h and 3 km/h b. 10 km/h and 2 km/h
c. 7 km/h and 10 km/h d. None of these

Sol. b. Let the speed of the sailor in still water = x km/h and speed of the current (stream) = y km/h
Then, speed of the sailor downstream = $(x + y)$ km/h and speed of the sailor upstream = $(x - y)$ km/h

$$\therefore \text{Time to travel 8 km downstream} = 40 \text{ min} = \frac{40}{60} \text{ h} = \frac{2}{3} \text{ h}$$

$$\Rightarrow \frac{8}{x+y} = \frac{2}{3} \Rightarrow x+y = \frac{24}{2} \quad \left[\because \frac{\text{distance}}{\text{speed}} = \text{time} \right]$$

$$\Rightarrow x+y = 12 \quad \dots(i)$$

$$\text{and time to return} = 1 \text{ h} \Rightarrow \frac{8}{x-y} = 1$$

$$\Rightarrow x-y = 8 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get $x = 10$ and $y = 2$

Hence, speed of sailor in still water = 10 km/h

Speed of current (stream) = 2 km/h

Rule 10 Let the speed of boat in downstream = u km/h and speed of boat in upstream = v km/h, then

- (i) Speed of boat in still water = $\frac{1}{2}(u+v)$ km/h
(ii) Speed of stream (current) = $\frac{1}{2}(u-v)$ km/h

EXAMPLE 13. The speed of boat upstream and speed of boat downstream are 7 km/h and 13 km/h, respectively. Then, the speed of stream and speed of boat in still water is

- a. 10 km/h and 3 km/h b. 15 km/h and 9 km/h
c. 20 km/h and 6 km/h d. 40 km/h and 12 km/h

Sol. a. Let the speed of boat in downstream and upstream be u km/h and v km/h, respectively.

Given, $v = 7$ km/h and $u = 13$ km/h

$$\text{Speed of boat in still water} = \frac{1}{2}(7+13) = \frac{1}{2}(20) = 10 \text{ km/h}$$

$$\text{Speed of the stream} = \frac{1}{2}(13-7) = \frac{1}{2}(6) = 3 \text{ km/h}$$

RACES

A contest of speed in running, driving, riding, sailing or rowing over a specified distance is called a race.

Suppose A and B are two contestants in a race. If A beats B by x m and length of the track is d m.

Then, Distance travelled by B when A finishes the race

$$= d - x \text{ and } \frac{\text{Speed of } A}{\text{Speed of } B} = \frac{d}{d-x}$$

Note (i) A gives B a start of x m means when A starts at starting point, B starts x m ahead from starting point.

(ii) A gives B a start of t s means A starts t s after B starts from the same point.

EXAMPLE 14. In a 200 m race, A can beat B by 50 m and B can beat C by 8 m. In the same race, A can beat C by what distance?

- a. 60 m b. 72 m c. 56 m d. 66 m

Sol. c. A can beat B by 50 m.

$$\therefore \frac{\text{speed of } A}{\text{speed of } B} = \frac{200}{200-50} = \frac{200}{150}$$

Also, B can beat C by 8 m,

$$\therefore \frac{\text{speed of } B}{\text{speed of } C} = \frac{200}{200-8} = \frac{200}{192}$$

$$\begin{aligned} \text{Now, } \frac{\text{speed of } A}{\text{speed of } C} &= \frac{\text{speed of } B}{\text{speed of } C} \times \frac{\text{speed of } A}{\text{speed of } B} \\ &= \frac{200}{150} \times \frac{200}{192} = \frac{200}{144} \end{aligned}$$

So, A beats C by $(200 - 144) = 56$ m